



Big Bang AR app

This app by CERN lets you visualise what happened during and after the Big Bang in your living room. <https://dgit.in/dec19-69>



Infinite yardstick

Explore what it would be like to have a universe sized scale, and go about measuring everything. <https://dgit.in/dec19-70>

There were annihilation events when leptons interacted with antileptons too. About ten seconds after the Big Bang, the photons popped into existence. Photons are their own antiparticles.

Because of mass energy equivalence, the high energy photons could then convert or decay into particles with mass, again in matter and antimatter pairs. During this time, if all the matter created had interacted with the antimatter, then all that would have been left of the universe after ten seconds would have been energy, in the form of gamma rays, which would result in an empty universe after a series of decays. The Big Bang was extremely stingy when it came to the business of creation.

However, a small fraction of the particles survived - only about one in every 10 billion that came into existence. In other words, the entire universe that we inhabit is made up of only 0.000000001 percent of the particles created after the Big Bang. Theoretically, equal amounts of matter and antimatter should have survived the creation and carnage in the first few seconds of creation. However, that is simply not the case. As far as we know, the universe around us contains far more matter than antimatter. Scientists have absolutely no clue why this is the case. Something must have happened in the very early universe, that resulted in more matter than antimatter surviving. The problem is known as baryon asymmetry. The Standard Model or the Theory of General Relativity both fail to explain why there is more matter than antimatter in the universe. Researchers at high energy particle physics facilities around the world are trying to solve this cosmic puzzle.

One of the simplest proposed explanations is that there may be regions of the universe where there are entire galaxies, star systems, planets and possibly even inhabitants made up of antimatter instead of matter. As only photons could possibly make the trip between these antimatter zones to Earth, to anyone on our planet, the antimatter galaxies would be indistinguishable from galaxies made up of matter. The photons would be the same in both

the zones. However, the boundaries between the realms of matter and antimatter, would constantly be marked by annihilation events, leading to flashes that could be detected from the Earth. The density of matter in the intergalactic medium, or the space between two galaxies is expected to be about one atom for every cubic metre of space. Based on this density, it is possible to calculate the frequency of such annihilation events, and check for gamma ray luminance in these boundaries. After scanning the skies for over 30 years, researchers have been unable to find a boundary region between zones of the universe where the two different types of matter dominate. This in turn allows us to calculate that any such region would have to be so far away, that we can never see it. Scientists now consider it unlikely that we will find a region of space within the visible universe, that is made up of antimatter galaxies. Everything we can see in the sky, is made up primarily of matter, and not antimatter.

The hearts of stars are furnaces that cook up the matter that we see all around us. The elements created after the Big Bang were mostly hydrogen and helium. The earliest stars were extremely massive, and were responsible for the formation of heavier elements in the periodic table, up to Iron. These stars were known as Population III stars, and none of these have been observed yet. The next generation, known as Population II stars, created the heavier elements, but were themselves poor in metallic content. The Population I stars are the youngest, metal rich stars such as the Sun and most of the other stars in the Milky Way. Scientists have only ever observed Population I and II stars. When the difference between metallic and low metallic stars were first discovered, the Sun like high metal stars were classified as Population I and the low metal stars were called Population II. The designation stuck. The Population III stars are actually the oldest, first generation stars. As the stars cooked up the heavier elements, they converted an increasing number of protons to neutrons. However, throughout the process the

total number of protons and neutrons, or baryons were conserved. While some processes can create pairs of baryons and antibaryons, the total number of baryons in the universe remains constant, and has been constant since the creation of the universe. This constant is known as the baryon number. Each baryon is made up of three quarks, which come in six flavours. This allows us to calculate the baryon number - which is one third of all the quarks in the universe. The baryon number, may help account for the excess of matter. To explain the baryon asymmetry, we would have to investigate the universe at the smallest scales, and uncover the secrets of quarks.

Andrei Sakharov a Russian nuclear physicist laid the foundation for future research in an explanation for baryon asymmetry in 1967. He outlined three basic conditions that were necessary for there being more matter in the universe. Sakharov reasoned that any explanation for baryon asymmetry must allow for some process that can change the baryon number of the universe. Scientists have not observed any such processes yet. The second condition was that there was some fundamental principle in the universe, that favoured the creation of matter over antimatter. While humans can be mostly glad that there is something instead of nothing, it can be a little scary to imagine the galaxies and worlds of antimatter that simply do not exist, just because the cosmos has a fundamental bias towards matter. The third condition is that the theoretical process that changes the baryon number of the universe, has to be out of thermal equilibrium. If heat could be exchanged between the particles interacting in the process, then the number of baryons and antibaryons would be evened out.

Sakharov has provided scientists with a handy roadmap for the explanation. Any resolution for baryon asymmetry would have to satisfy the three conditions laid out by Sakharov. While scientists have some theories, these can only be tested in the next generation of more powerful particle accelerators. 